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8.3.1 A-conjugate directions

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this set that only consists of these lines it sure would be nice, if instead it minimized over the span of all of the search directions. Indeed if these search directions were linearly independent, then once k is equal to $n-1$, in other words, x super n , $x^{(n)}$, would then be the minimum over the span of n linearly independent vectors. If our matrix is n by n that means that x , that the function is minimized over x in all of $\mathbb{R}^n$. And what

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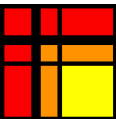
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Reading assignment

1/1 point (graded)



Read Unit 8.3.1 of the notes. [\[LINK\]](#)

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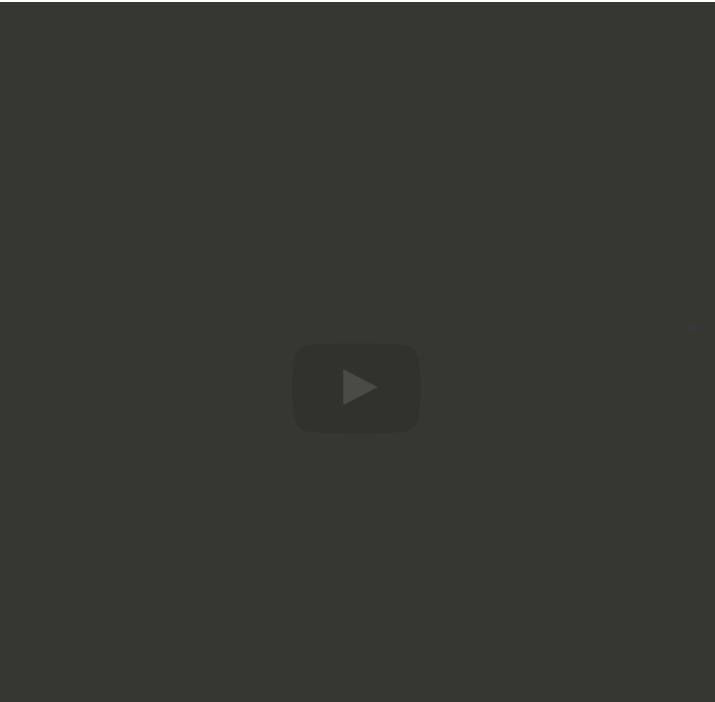
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tor where the
solution is. Okay? So
with that, we can take
our expression right
here and just
put it there again.
And what we then
notice is that, hey,
instead of thinking
of the span of all of
these vectors, let's
make those vectors
into the
columns of a matrix.
And then talking
about all linear
combinations of
those
columns is the same
as saying take
vectors y of the
appropriate size and
look at all products of

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Homework 8.3.1.1

1/1 point (graded)
Let $p^{(k)}$ be a new search direction that is linearly independent of the columns of $P^{(k-1)}$, which themselves are linearly independent. Show that

$$\begin{aligned} \min_{x \in \text{Span}(p^{(0)}, \dots, p^{(k-1)}, p^{(k)})} f(x) &= \min_y f(P^{(k)} y) \\ &= \min_y \left[\frac{1}{2} y_0^T P^{(k-1)T} A P^{(k-1)} y_0 - y_0^T P^{(k-1)T} b \right. \\ &\quad \left. + \psi_1 y_0^T P^{(k-1)T} A p^{(k)} + \frac{1}{2} \psi_1^2 p^{(k)T} A p^{(k)} - \psi_1 p^{(k)T} b \right], \end{aligned}$$

where $y = \begin{pmatrix} y_0 \\ \psi_1 \end{pmatrix} \in \mathbb{R}^{k+1}$.

☒ done

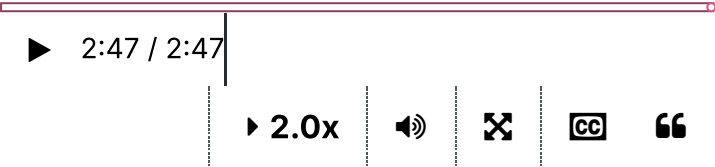


Solution:

For a complete solution see the [notes](#).

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right properties. And what is that property then. Well, if only this were true about our search directions -- let's write that down -- if only $P^{(k-1)}$ times A transpose here times A times little $p^{(k)}$ were equal to 0, then we end up in this very nice situation. So here now we have a property that we want from our search directions. Now if we took A away, then this would say that our next search direction is orthogonal to all of

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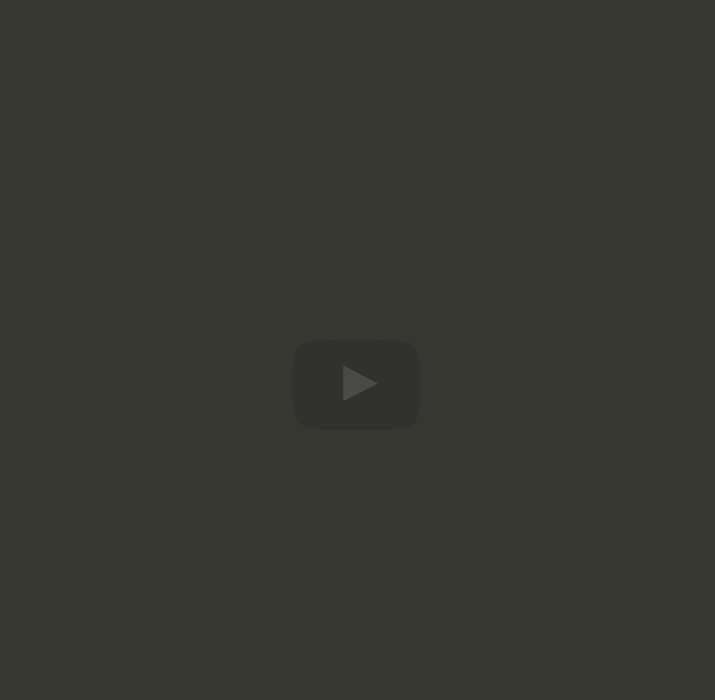
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one more term to $x^{(k)}$ -- this is $x^{(k)}$ -- in order to come up with our next approximation for our solution. So now we can either say, "Look! We can actually use this expression to compute our ψ_1 or you can recognize that this whole argument was all about how should we be picking our search directions." And if that is what it's all about, then all we do is say, "Okay, we're going to pick our search

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Homework 8.3.1.2

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be SPD.

ALWAYS/SOMETIMES/NEVER: The columns of $\mathbf{P} \in \mathbb{R}^{n \times k}$ are A-conjugate if and only if $\mathbf{P}^T \mathbf{A} \mathbf{P} = \mathbf{D}$ where $\mathbf{D}$ is diagonal and has positive values on its diagonal.

ALWAYS

 Answer: ALWAYS

Solution: ALWAYS

Now prove it!

For a complete solution, see the [notes](#).

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 Answers are displayed within the problem

Homework 8.3.1.3

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be SPD and the columns of $\mathbf{P} \in \mathbb{R}^{n \times k}$ be A-conjugate.

ALWAYS/SOMETIMES/NEVER: The columns of $\mathbf{P}$ are linearly independent.

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